

US Department of Agriculture (USDA)



Statement of Work

Barrens Rare Insect Survey

Hoosier National Forest

July 23, 2026

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Project

Barrens Rare Insect Surveys

General Information

1.0 Scope of Work

Work shall consist of surveying three Barrens ecosystems on the Hoosier National Forest, Tell City District for remnant-dependent grassland insect species whose life cycle requirements limit their distribution to barrens plant communities and are rarely found in the surrounding more highly disturbed forests. The goal of this project is to determine if: 1) insect species surveyed in these areas previously are still present, 2) if present, evaluate how their abundances have changed since 2002, 3) provide explanations for changes seen and 4) provide restoration recommendations to boost and expand populations of these species.

All three barrens areas are in Perry County, Indiana. Boone Creek Barrens is ~700 acres and is located near Magnet, IN (38.14353, -86.47765). Clover Lick Barrens is ~1658 acres and is located near Derby, IN (38.06226, -86.57190). Harding Flats Barrens is ~782 acres and located near Gerald, IN (37.99410, -86.59623).

A total of eight, three-day sampling trips will occur on the Hoosier National Forest, for a total of 24 sampling days. Each barrens site should be visited at least once a day during these sampling trips, but the duration of each survey can be dependent upon the apparent abundance of the target insect groups.

Performance Period: 04/05/2027 – 03/31/2028. Sampling must occur from April – October 2027 and the final report is due by March 2028.

2.0 Background

The USDA-Forest Service Hoosier National Forest proposes to contract for technical scientific services to complete rare insect monitoring. Its purpose is to obtain biological information on the status of remnant-dependent species and to utilize these survey counts to provide the Government information with which management determinations for natural resources will be made.

These barrens sites have been surveyed 2-3 times previously. Boone Creek and Clover Lick were sampled in 2002, 2016, and 2020. Harding Flats was surveyed in 2002 and 2016. Since each site was last surveyed multiple prescribed fire and tree thinning have occurred within each area to restore barrens habitats. Upon award, survey results, maps, and restoration histories will be provided to the contractor. Current surveys are being advertised to continue monitoring efforts and develop protocols and plans to increase restoration and revegetation.

Contractor Requirements

3.0 Technical Requirements / Tasks

General Information

- The Contractor shall provide all expertise, labor, supervision, services, materials, equipment, tools, supplies, transportation, worker personal protective equipment (PPE), and incidentals necessary to perform the required work, for which the desired outcome is in strict accordance with the standards in the contract.
- The Contractor shall perform and accomplish the work in accordance with all applicable federal, state, and local laws, regulations, and ordinances.
- Where appropriate, vehicle or ATV/UTV/OHV is permissible down access routes and inside designated openings.
- The project area shall be accessed by State, County or Forest roads. The Forest Service assumes no obligation to do special maintenance to keep roads open. Stands are accessible by some combination of 2- or 4-wheel drive vehicle and foot travel.
- The Contractor shall progressively accomplish the work at a rate that will result in completion of the work within the time specified as determined by the final accepted work schedule for the awarded contract.
- Before implementation, the project leader will meet the contractor on site.
- Contractor shall notify COR no less than 48 hours prior to start of contract work.
- In cases where stands are behind locked gates, authorization for limited motorized use may be granted. All requests for motorized travel of closed roads must be submitted in writing to the COR for specific roads and specific periods of time. The Contractor will be responsible to ensure that gates remain closed and locked at all times and will be subject to all other restrictions of administrative use. The Contractor shall be permitted to drive from the locked gate to the stand with only the minimum number of vehicles necessary for efficient transport of the crew. No access to Contractors or employees will be allowed behind locked gates after periods of work. Any misuse of this authorization may result in revocation of the permission

Target Insect Taxa

Species targeted are those identified as remnant-dependent grassland species that are both fire dependent AND fire sensitive. Sampling efforts are to be focused primarily on insects in the order Lepidoptera (including butterflies, skippers, and moths),

Orthoptera (specifically within the family Acrididae), Hymenoptera (specifically the Andrenidae, Apidae, Colletidae, Halictidae, and Megachilidae families) and the following families in Homoptera: Caliscelidae, Cicadidae, Cixiidae, Delphacidae, and Dictypharidae. Other insects that are captured as by-catch or opportunistically should also be included. A list of target species will be provided to the contractor upon award.

Sampling Periods and Duration

A total of eight, three-day sampling trips will occur on the Hoosier National Forest, for a total of 24 sampling days. Each barrens site should be visited at least once a day during these sampling trips, but the duration of each survey can be dependent upon the apparent abundance of the target insect groups. Individual sampling trips will be spread from April – October 2027, corresponding to regional flight periods for the target taxa (regional flight periods for specific target insects and specification for which insects to target during each trip will be provided to the contractor upon award). Specifically, these sampling trips will be within the following timeframes in 2027:

- Trip 1: April 10 – 25
- Trip 2: May 5 – 15
- Trip 3: May 25 – June 10
- Trip 4: June 25 – July 10
- Trip 5: July 15 – 30
- Trip 6: August 10 – 30
- Trip 7: September 1 – 15
- Trip 8: September 25 – October 15

Sampling Methodology

Methodology will be repeated from the previous surveys in 2002, 2016, and 2020 to be consistent with surveys during those timeframes. Overviews and details are provided below and full guidelines, maps, and reports from previous year's surveys will be provided to the contractor upon award.

Walkthrough visual surveys and aerial net surveys will be completed for bees, butterflies, and skippers during daylight hours, whenever the weather permits (>60 degrees Fahrenheit, no rain).

Sweep net sampling will be completed for Homoptera and Orthoptera species, using 15-inch diameter, muslin nets. Samples will be concentrated in areas dominated by herbaceous vegetation consisting of grasses, sedges and forbs. Samples can vary from 100 to 500 sweeps depending on size of area. Sweeps to be taken in 120-degree arcs at ground level with the net face perpendicular to surface of the ground in front of surveyor. Sweep net samples should be collected late in the day (1600 - 1800 CST), to coincide with peak activity for many target species.

Ultraviolet (UV) light trap surveys will be used to collect night-flying target species. UV light traps will be run during each of the 2-nighttime periods over each sampling period.

The dominant methodology used each sampling trip will vary depending on the time of year the trip is being completed, as the methodology used during each trip will be adjusted to best capture the most active target species, based off regional flight periods and flowering vegetation (regional flight periods for specific target insects and specification for which insects to target during each trip will be provided to the contractor upon award).

All notable (i.e., remnant-dependent or otherwise of conservation concern) individuals captured will be noted and accurately identified. Collected samples will be stored on ice until the time which they can be sorted and identified.

Report of Findings

A report on results is required to compare previous survey results to results of this contract. A table showing all species identified and their presence or absence in each barrens area is required. Focus is to be on the remnant-dependent insects and compare changes in individual species' presence and abundance from each survey year. Likewise, the report shall document absence of remnant-dependent insects previously known in the areas but not found during the current survey. If species are found missing (e.g. lost or not yet recovered), provide recommendations for future management that would lessen pressure on these species and allow for their recovery after prescribed burns that would not endanger their long-term survival. Previous reports and survey results will be provided to the contractor to use as a template upon award. Report must be submitted electronically no later than March 31, 2028.

Specifically, the report should include:

- Background
 - Brief background on barrens remnant-dependent species, fire dependence and fire sensitivity, and the three barrens ecosystems.
 - Background information on any newly identified barrens remnant-dependent insects and ecosystem management impacts (fire, cutting, etc). Information on this is not needed for previously identified insects unless substantial changes regarding their conservation status have changed.
 - Include description for prescribed burning that occurred at each two sites (this information will be provided by the U.S. Forest Service).
- Methodology
 - Detail the methodology used to complete the surveys.
- Results and Discussion
 - Comparison of barren remnant-dependent insects found in across each of the four survey years, unique for each site and combined.
 - This would include the number of different species AND observed changes in abundance. Note any missing remnant-dependent species and discuss why they are missing.
 - Any new, notable insects or barrens remnant-dependent insects found.
 - Discussion of changes seen between individual insect species and groups found across different surveys, and potential reasons for these changes.
- Recommendations
 - Recommend future barrens habitat restoration and management.
 - Recommendations on any mitigations needed for maintaining viable populations of rare and barrens remnant-dependent insects.
 - Recommendations on expanding insect populations and abundances, both by increasing numbers in current habitats and creating new habitats.
 - Unique recommendations for each barrens area should be made, as well as overlapping recommendations for all barrens.
- Maps of surveyed areas that highlight different quality habitats.
- Results tables
 - Tables should include (but are not limited to) total species found in current surveys, differences from previous surveys, and new species observed.
 - Tables should include both presence/absence and abundance results.
 - Tables should be made for each specific barrens area and all three combined.
- Photographs of both sampling methods and collected insects.
- Bibliography

Inspections

Inspections will be completed in real-time. The COR will join the contractor for at least 1 day per sampling trip to ensure all guidelines are being met.

Invoicing

Invoicing is tied to successful inspection of completed work. The Contractor may only invoice after the COR has inspected the line item(s) work and the work is determined to have met the AQL. This applies to both interim and final payments. All invoices will be submitted via U.S. Treasury's Invoice Processing Platform (IPP).

4.0 Government Furnished

The Government shall provide the Contractor the following listed materials, supplies, property or services, for use in the performance of this contract, hereinafter referred to as Government Furnished Property. This property shall be used and maintained by the Contractor in accordance with the provisions of the "Government Furnished Property." FAR clause, (Reference Clause 52.245-1).

ITEM	DESCRIPTION	WHERE, WHEN, HOW
Project area maps	Maps depicting all area boundaries for this project.	Provided with task order
Contractor keys	Some stands are behind Forest road closures (locked gated) and will require keys to access.	Pre-work meeting

ITEM	DESCRIPTION	WHERE, WHEN, HOW
Previous reports, results, methods, and maps	Reports, maps, results, and protocols needed to align the current surveys with previously completed surveys.	Provided electronically upon award
Target insect lists	Lists of target species and their activity periods to best design sampling methodologies and dates.	Provided electronically upon award

5.0 Deliverables / Schedule

Deliverables:

- Completion of eight, three-day sampling trips on three difference barrens areas.
- Report of findings and recommendations.
- Upon award, the Contractor shall prepare a work plan and schedule that demonstrates work will be completed within the performance period described in individual orders and over each sampling period. The Contractor shall provide a general plan with the technical proposal. A supplementary plan and schedules will be provided after issuance of an award, but before start of work.
 - The Contractor shall advise the Contracting Officer or COR of any periods that the Contractor will not be working that is not specified in the accepted Work Plan.
 - The Contractor may update the work plan when work is in progress for the Contracting Officer's approval.

Schedule:

- The Contractor shall progressively accomplish the work at a rate that will result in completion of the work within the time specified as determined by the final accepted work schedule for the awarded contract.
- Work must be done during the specific sampling periods defined above.
- The report of findings must be completed after all surveys are completed and by **March 31, 2028**.

Performance period: 04/05/2027 – 03/31/2028

Contract Line-Item No. – Work Requirement	Task / Critical Sub-tasks	Inspection requirements	*Acceptable Quality Level (AQL)	Monitoring Method
Non-line item: Final report	Report of findings	COR will evaluate the report of findings to ensure all sections and information required (listed above in section 3) are included.	All sections are included. Information included matches what was required. Information can be used by Hoosier National Forest staff to monitor insect populations and achieve restoration actions.	Visual

Contract Line- Item No. – Work Requirement	Task / Critical Sub-tasks	Inspection requirements	*Acceptable Quality Level (AQL)	Monitoring Method
Line Item: Boone Creek Surveys	Completion of 8, 3-day surveys. Surveys completed in the appropriate date ranges. Appropriate methodology is used for target taxa.	COR will inspect 1 days’ worth of surveys during each sampling trip.	Surveys are being completed when required. All survey methods are being utilized and altered based on local conditions. Insects are being accurately identified or stored for future identification.	Visual
Line Item: Clover Lick Surveys	Completion of 8, 3-day surveys. Surveys completed in the appropriate date ranges. Appropriate methodology is used for target taxa.	COR will inspect 1 days’ worth of surveys during each sampling trip.	Surveys are being completed when required. All survey methods are being utilized and altered based on local conditions. Insects are being accurately identified or stored for future identification.	Visual
Line Item: Harding Flats Surveys	Completion of 8, 3-day surveys. Surveys completed in the appropriate date ranges. Appropriate methodology is used for target taxa.	COR will inspect 1 days’ worth of surveys during each sampling trip.	Surveys are being completed when required. All survey methods are being utilized and altered based on local conditions. Insects are being accurately identified or stored for future identification.	Visual

*Meeting the AQL for Line Items: Contractor(s) must re-compete plots at their expense upon failure to meet AQL values at the final inspection for each contract line item where the work performed failed to reach the minimum acceptable target ratio.

6.0 Travel

The contractor will be responsible for their own travel expenses and for traveling to and from the sampling areas for each completed survey.

7.0 Contractor’s Key Personnel

Successful performance of the Barrens Rare Insect Surveys requires personnel with strong scientific field experience, familiarity with insect sampling methodologies, knowledge of barrens ecosystems, and the ability to produce high-quality technical reporting. The following positions are designated as **Key Personnel** for this contract because they directly ensure that sampling, identification, data integrity, and reporting meet the standards outlined in the SOW. One person may fill more than one role, but each role must be filled by at least one person.

1. Project Lead

Role: Responsible for overall execution of the project, coordination and/or completion of fieldwork, coordination with the COR, development of survey methodology refinements, quality control of species identifications, data synthesis, and preparation of the final report.

Required Qualifications:

- Advanced experience conducting insect surveys, specifically Lepidoptera, Orthoptera, Hymenoptera, and Homoptera.
- Demonstrated capability in planning and conducting multi-day field sampling efforts.
- Experience with fire-dependent and fire-sensitive insect species, barrens ecosystems, or similar natural communities.
- Past performance on comparable rare species monitoring for similar habitats

2. Entomologist / Insect Taxonomist

Role: Completes daily sampling at all three barrens sites, performs visual, sweep net, and UV trap sampling, identifies remnant-dependent insects, and manages sample preservation and labeling.

Required Qualifications:

- Strong taxonomic skills across Lepidoptera, Hymenoptera families listed (Andrenidae, Apidae, Colletidae, Halictidae, Megachilidae), Orthoptera, and targeted Homoptera families.
- Demonstrated experience in netting methodology, sweep sampling, and nocturnal UV trapping.
- Ability to accurately identify notable and conservation-concern species.

Submission of resumes is allowed but not required. However, all pertinent information must be filled out in the evaluation criteria.

Submissions that do not include the evaluation criteria will not be able to be fully evaluated.

Definitions:

1. Contracting Officer (CO): A Forest Service employee with the authority to: enter into, administer, and terminate contracts, as well as issue determinations regarding such matters as claims.
2. Contracting Officer's Representative (COR): A Forest Service employee that is appointed and officially designated by the Contracting Officer to assist the Contracting Officer with the technical matters related to contract administration. The COR will assist the Contracting Officer with surveillance of contractor performance.
3. Forest Service Inspector: A Forest Service employee designated by the Contracting Officer or COR to perform surveillance of contractor performance.
4. Acceptable Quality Level (AQL): A suitable level of conformance based on an established percentage of defects.
5. Quality Assurance: Those actions taken by the Forest Service to assure services meet specified requirements.
6. Quality Control: Those actions taken by a Contractor to control the performance of services so that they meet the specified requirements.